Event-Based Structured-Light 3D Scanner: Static Reconstruction and Evaluation

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Abstract—We built a 3D scanner that projects a thin red laser line onto an object from a motor-driven mount and watches it with a camera that records both ordinary images and per-pixel brightness changes (events). The team designed and printed the mounts and wrote the control and processing software. Holding the laser still at 20 calibrated positions, we reconstructed partial 3D profiles of a shoe: 960 points from images and 494 from events. On a calibration board pose kept out of the fit, the equal-position root-mean-square depth error was 8.63 mm, ranging from 0.88 to 17.03 mm by position. On rows both methods found, events gave 8.07 mm and images 8.42 mm, so neither was clearly better. With the laser dimmed by a filter, the image method still found the stripe in 62–64 rows while the event method found 0–1. A timing prediction test left residuals up to 47.55 ms, so continuous reconstruction remains unvalidated. Independent object accuracy is untested. Code, retained results and a runnable coverage example are public; the full recording archive is not yet released.

Index Terms—Structured light, event cameras, triangulation, calibration, experimental validation.

1 Introduction

A MOVING projected line makes surface shape observable through camera–projector triangulation. An event sensor is attractive because stripe transitions can produce temporally local observations, while ordinary frames also sample the background. The proposed research question was whether calibrated depth could be recovered from a smoothly swept weak line, and how error and coverage change with sweep speed and optical attenuation [1]. Those advantages are hypotheses, not consequences of choosing an event camera.

We implement the proposal's *stop-and-pulse fallback*: held commands and bounded pulses yield grayscale/event candidates for fixed-plane triangulation. Results cover partial profiles, reserved-board errors, repeatability, image coverage and one unknown-filter test. Continuous reconstruction remains unvalidated (Section 6).

1.1 Contribution beyond the laboratories

The team designed and printed the base, camera/servo mounts, rotating projector cradle and pedestal, and integrated the DAVIS240C, feedback controller and processing software (Fig. 1). Active geometric depth measurement and actuator integration extend the lab kit beyond classical imaging, relighting, coded apertures and lensless reconstruction [2], [3], [4], [5]. Fixed settings, measured calibration, explicit baselines and retained failures carry those laboratory methods into this physical scanner.

2 Related Work

MC3D combines motion-contrast sensing with scanned illumination to exploit sparse temporal measurements [6]. It motivates our scanner, but its reported capabilities cannot be transferred to our servo implementation. ESL uses spatio-temporal consistency to address event matching and timing effects in a raster-scanned projector system [7]. We do not implement its optimisation or claim equivalent performance. The event-vision survey of Gallego *et al.* [8] provides the broader sensor/algorithm context. Our camera calibration follows the planar-target approach associated with Zhang [9], followed by a documented corner-detection audit. The project contribution is a course-scale physical implementation and evaluation, not novelty over these papers.



Fig. 1. CAD design (left), team-printed mechanics (centre) and assembled scanner (right). CAD is intended geometry; photographs document the physical contribution. These images do not establish as-built dimensions, optical power or depth accuracy.

3 System and Reconstruction Method

3.1 Physical and software implementation

The DAVIS240C records 240×180 grayscale active-pixel-sensor (APS) frames, events and inertial samples. A feedback servo rotates the line projector under Uno-compatible firmware v1.7. One Chrome dashboard owns serial control and one service owns the DAVIS. Unique operation IDs link recordings, sidecars, controller messages and feedback; uncertain operations are inspected before retrying. Fig. 3 shows the implemented static workflow.

• *MECH5720, The University of Sydney, 2026.*

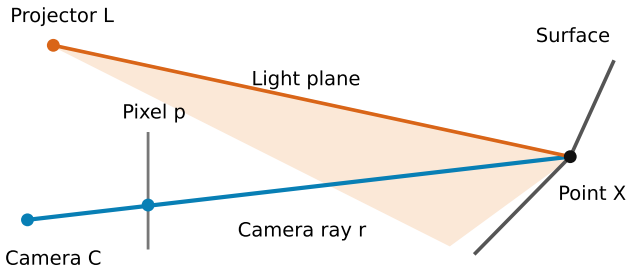


Fig. 2. Conceptual ray–plane intersection, not to scale. The camera ray through the stripe pixel intersects a static fitted light plane. This geometry does not establish timing during motion.

The firmware accepts USB text commands and uses a consistent approach direction to reduce backlash. It records controller-local feedback and laser markers in microseconds. Firmware checks require arming, an interlock input and a configured PWM window before emission; manual operation has a 60 s timeout. These software checks do not independently verify physical containment or calibrated angular limits. Settle timeouts can warn rather than abort; warnings are retained.

The implemented observation protocol holds the current command for 3 s in darkness, emits for 1 s, records 0.5 s after extinction, and ends off/disarmed with the servo released. Native camera recordings span 12 s, including setup margins. Feedback variation is advisory in observation mode, and a completed trace is not proof of a settled optical plane. Host controls are not safety-rated, and no feedback-to-degree conversion is assumed.

The pitch specified an Uno R4 and transistor-switched Class 1 module. The recorded build instead uses an R3-type controller, an MXD1230 650 nm line module supplied from D7 and a feedback servo identified in team notes as DFRobot DMS-MG90-A. The team reported Class 3 operation and verbal teaching-team advice; neither optical class/output nor the electrical design was independently verified in this study. Mounts fix the camera–projector arrangement, but the as-built baseline, final lens configuration and physical datums were not surveyed. CAD dimensions are not ground truth.

3.2 Camera rays and static light planes

Let $\mathbf{p} = (x, y)$ be a retained stripe centre. Undistortion using the camera matrix K and distortion parameters D gives ray $\mathbf{r} = (x_n, y_n, 1)^\mathsf{T}$. For a calibrated static command u, ray–plane intersection (Fig. 2) gives

$$\mathbf{n}_u^\mathsf{T}\mathbf{X} + d_u = 0, \qquad \mathbf{X} = -\frac{d_u}{\mathbf{n}_u^\mathsf{T}\mathbf{r}}\mathbf{r}. \tag{1}$$

We reject near-parallel, nonfinite or nonpositive-depth intersections. Distances inherit the target's reported millimetre scale; printed pitch and flatness were not independently verified. Here u indexes a PWM command, not measured shaft angle. Continuous operation would require a validated camera-time-to-feedback/plane mapping, which is not supplied by matching timestamp ranges.

The intrinsic calibration uses a 9×6 inner-corner target with a reported 18 mm square pitch. A 60 s recording contains 821 APS frames. Twenty-five candidate views were re-detected consistently with a sector-based detector at $4\times$ resolution and cross-checked at $2\times$. Three training candidates were excluded for cross-scale disagreement or grid anomalies, and the original data and reasons were retained. Seventeen training views and five reserved views were used for the final audit. Reserved views had already been inspected during diagnosis, so they are not an untouched test set.

For light-plane calibration, detected stripe rays intersect each posed board to produce 3D lines. Multiple lines at one command define a plane. Poses A–C fit the model, and D is excluded and used for validation (Fig. 4). Twenty planes from 1569 to 1647 μs were fitted: 18 use three board poses and two use two. The insufficiently supported 1651 μs command is absent. The model was frozen before D evaluation and hash-checked afterwards. No D alignment, refit or bias correction was applied.

3.3 APS and event stripe candidates

The APS detector subtracts the first three-frame median, selects frames whose global positive-change signal exceeds half its peak, and retains at least five lit frames. On rows 20–105, candidate pixels require change > 20 and intensity > 40 digital levels. Contiguous row groups must be 1–12 pixels wide. The strongest group is rejected if a runner-up exceeds 0.67 of its energy or any selected pixel is ≥ 250. A row median is retained when supported in at least half the selected lit frames.

The event detector uses no APS spatial stripe mask. In the same fixed image rows, 25 ms polarity-count peaks separated by the known 1 s pulse duration select candidate ON/OFF transitions. Primary windows are 50 ms per transition, with 20 and 100 ms as sensitivity settings. Pixels require at least three events and background-subtracted count at least two, and the same group-width/ambiguity screens are applied. ON and OFF row centres must agree within 3 pixels. This corroboration is a heuristic, not confirmed laser identity. Parameters were developed on saved shoe data, then applied unchanged to reserved-board and fresh repeat recordings. There is no untouched, manually labelled detector benchmark.

Both methods share recordings, camera model and planes. APS pools selected 35 ms frames; events use two 50 ms windows. Scene/geometry are matched, but temporal integration, photon budget and speed are not. Fig. 5 illustrates the two recorded outputs in a separate five-line demonstration; event accumulation is not the APS baseline.

4 Experimental Design and Metrics

4.1 Depth, agreement and repeated acquisitions

For the D board, an observation's ray intersects the posed reference board to give Z_{ref}. Depth residuals are $e = Z_{\mathrm{pred}} - Z_{\mathrm{ref}}$. We report per-command RMS and equal-command aggregation,

$$E = \sqrt{\frac{1}{M}\sum_{u=1}^{M}\frac{1}{N_u}\sum_{i} e_{ui}^2}. \tag{2}$$

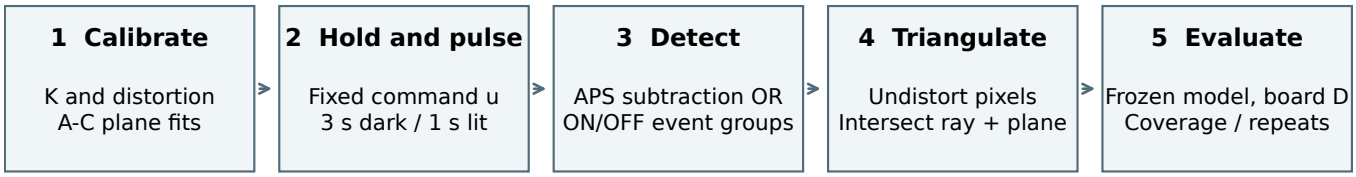


Fig. 3. Implemented acquisition-to-3D pipeline. Calibration is frozen before evaluation. Both detectors produce pixel candidates for the same ray–plane model; event timestamps do not supply a validated moving-plane model. Commands index fitted planes, not measured shaft angles.

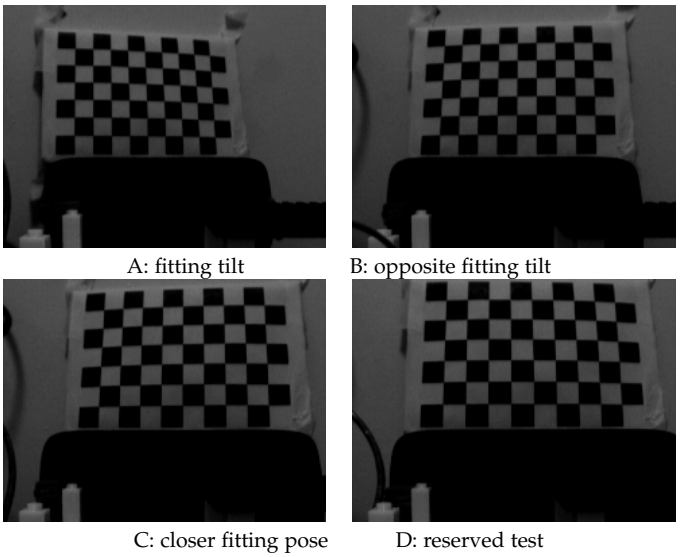


Fig. 4. Actual DAVIS board reference images. Poses A–C fit each light plane, and D is held out for testing. The 18 mm square size is the reported print scale. Pitch, print distortion and flatness were not independently verified.

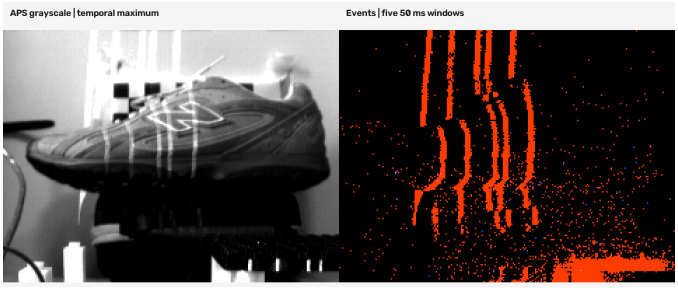


Fig. 5. Recorded five-line demonstration: ordinary APS grayscale temporal maximum (left) and 50 ms event-window accumulation (right). Accumulated activity is not confirmed laser detection. This separate display run is not the 20-position quantitative reconstruction.

The event/APS comparison uses only common command/row pairs inside the known inner-corner rectangle. This prevents a method's extra rows changing the comparison denominator, but may favour easy detections. The board reference shares K and the target scale, so it is not independent absolute ground truth. Shoe event–APS depth disagreement likewise measures agreement, not accuracy.

A later shoe placement defines a separate repeatability condition. Three acquisitions at each of five fitted commands were evaluated without alignment. For rows present in both methods and all three trials, precision is

$$S_u = \sqrt{\frac{1}{|R_u|}\sum_{y\in R_u}\frac{1}{n-1}\sum_{j=1}^{n}(Z_{uyj}-\bar{Z}_{uy})^2}, \quad n=3. \tag{3}$$

The independent units are acquisitions, not rows/events. An image row samples a ray, not a tracked material point. S_u combines mechanical, detector and scene variability. Common-row selection can suppress failures, so support counts are also retained.

4.2 Fixed-denominator image-space coverage

A manually outlined visible-shoe-body mask Ω in a laser-off photo contains 13,715 native pixels. It excludes background, pedestal and loose laces above the main body. Retained centres are rounded to native bins and deduplicated. With occupied set B, coverage is

$$C_{\text{image}} = 100\,|B\cap\Omega|/|\Omega|. \tag{4}$$

The same post-acquisition, approximate mask is used for both methods. Boundary perturbations are sensitivity checks, not confidence intervals. This metric is image-space sample occupancy, not physical 3D surface-area completeness or accuracy. No gap filling or mesh is counted as measurement.

4.3 Filter and optical timing diagnostics

The filter retry uses two recent unfiltered controls and three filtered acquisitions at 1626 μs, fixed 35 ms APS exposure and 1 s pulses. A fixed unfiltered-derived neighbourhood measures background-subtracted digital signal. The filter marking, transmission and camera radiometric response are unknown. Insertion leaves residual stripe displacement, so the digital ratio is a configuration diagnostic, not calibrated optical attenuation. Earlier disturbed trials are retained separately.

For timing, an affine relation was fitted to five first-pass ON markers from static pulses. Ten subsequent ON and all 15 OFF markers were reserved for prediction. Residuals include optical/polarity response and selection effects as well as clock-model error, so they are not isolated hardware latency or absolute timing uncertainty. Resets invalidate this session-specific mapping.

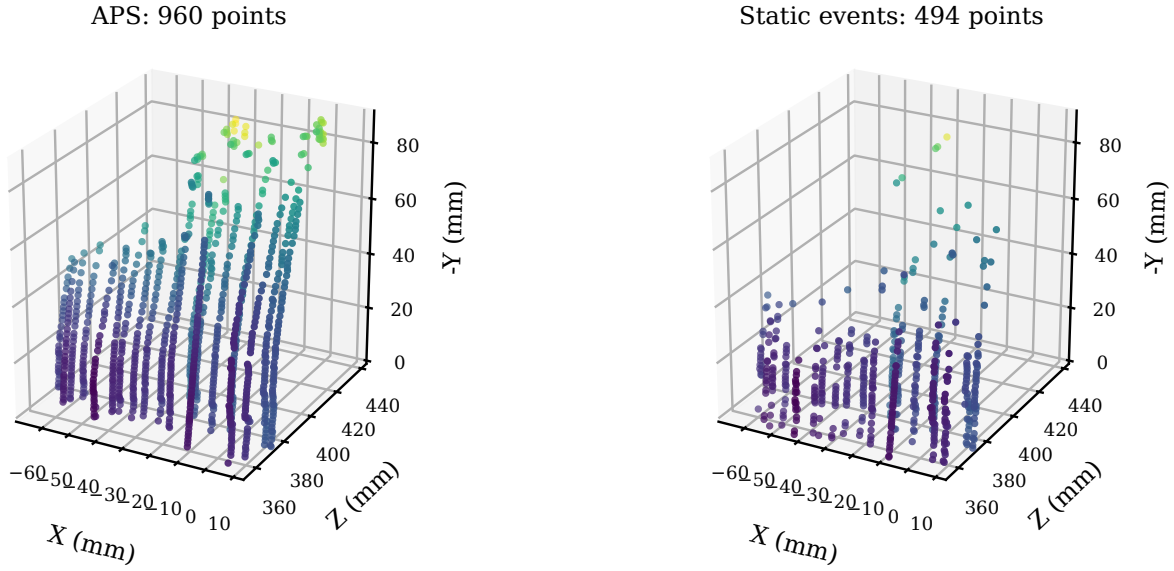


Fig. 6. Actual retained partial shoe point clouds on equal coordinate axes. Camera-coordinate millimetres inherit the target scale. No interpolated plane, surface mesh, test alignment or independent shoe accuracy is claimed.

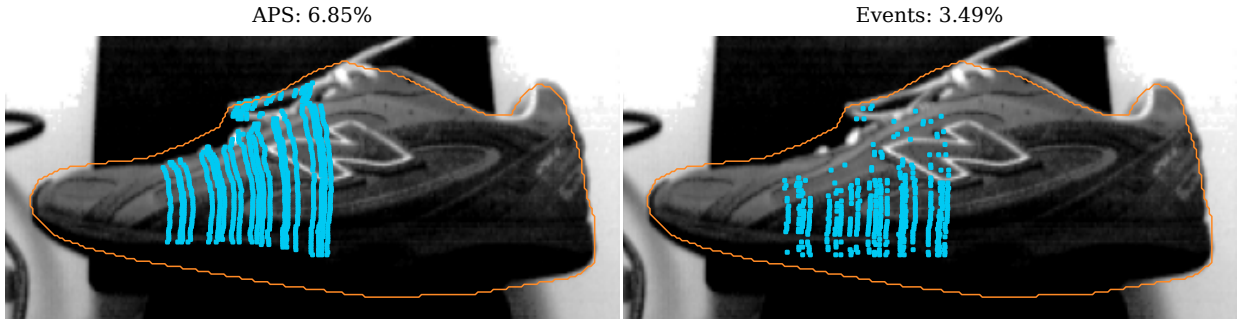


Fig. 7. Same fixed visible-body outline for both outputs. Cyan bins show retained samples, not filled surface. The denominator includes the toe, heel and lower body even where no profiles were reconstructed.

5 Results and Analysis

5.1 Camera and reserved-board calibration

The refined camera fit gives training RMS 0.2351 pixels, and five reserved per-view RMS values range from 0.1990 to 0.2965 pixels. The matrix is

$$K = \begin{bmatrix} 307.0201 & 0 & 115.7638 \\ 0 & 306.8715 & 104.0087 \\ 0 & 0 & 1 \end{bmatrix}. \quad (5)$$

The five distortion coefficients are approximately $(-0.28245, 0.55350, 0.003641, 0.001042, -0.89863)$. Leave-one-training-view-out focal ranges are 305.90–307.60 and 305.68–307.43 pixels. Fixing $k_3 = 0$ gives nearly identical training RMS 0.23515 pixels. Fig. 8 shows the reserved-view improvement after corner re-detection. These are sensitivity observations, not confidence intervals or millimetre depth validation.

The original APS-only reserved-D evaluation gives equal-command RMS 8.63 mm across 20 commands, with per-command RMS 0.88–17.03 mm. Restricting both methods to 339 common rows across 19 commands gives 8.42 mm for APS and 8.07 mm for events (Fig. 9). One command lacks common retained rows. The different denominator explains why the APS value differs from 8.63 mm. It is not an improved calibration. Position-dependent errors dominate any small aggregate difference, and no event advantage is established. This single reserved board pose supplies no independent trial-to-trial uncertainty for the headline depth RMS; correlated rows cannot substitute for repeats.

5.2 Partial shoe reconstruction and sampling

Twenty fitted positions yield 960 APS points and 494 event points at the primary window (Fig. 6). On 455 shared shoe rows, horizontal disagreement is 0.3385 pixels RMS and depth disagreement 1.6497 mm RMS. Both outputs share calibration errors, so close agreement does not prove object accuracy. Event windows 20/50/100 ms retain 359/494/532 points. All three settings retain zero corroborated candidates

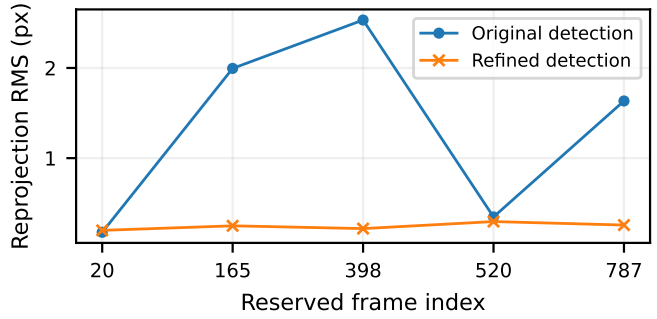


Fig. 8. Same five reserved images before and after uniform corner re-detection. Intrinsics exclude these views, but they were inspected during diagnosis; board poses are fitted with intrinsics fixed. Pixel reprojection errors do not validate metric depth.

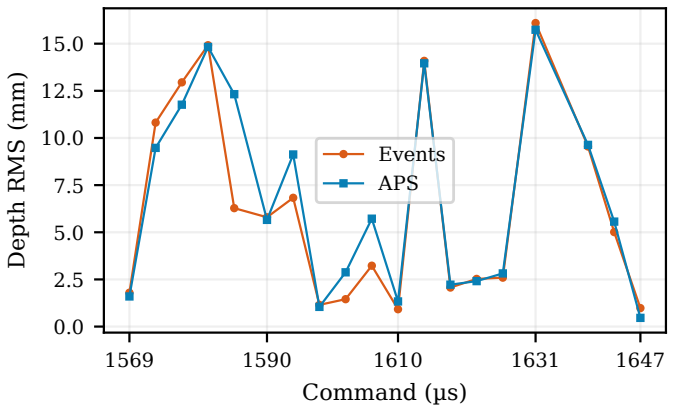


Fig. 9. Reserved D depth RMS on identical command/row support. Shared camera model and target scale, one board pose and no test alignment. Missing common support is not assigned zero error.

in 3,440 sampled laser-off command/row opportunities. These are internal controls, not a manually labelled detector false-positive rate.

Of the 960 APS centres, 955 lie in the fixed body outline and occupy 939 unique bins. All 494 event centres lie inside, occupying 478 bins. Full-body image coverage is therefore 6.85% and 3.49%, respectively (Fig. 7). One-pixel erosion/dilation changes these to 7.06/6.64% and 3.62/3.36%. This is a sparse partial scan, not full-shoe reconstruction. The historical 379-point/five-position scan used 10 ms APS exposure; a different 1,929-point/21-position preview used 3 ms and interpolated planes. Distance, model and crop also changed, so these are not controlled density or accuracy comparisons.

5.3 Fresh scene-stripe repeatability

Table 1 and Fig. 10 show no consistent winner across commands. APS retains more total rows with these settings, but common support contains only 8–21 rows per command. Before/after image phase correlation gives shifts $(-0.024, 0.031)$ pixels with response 0.9995, and no registration is applied. Results describe scene-stripe precision rather than isolated shoe accuracy. An initial command-selection error used 1627 μs, which has no fitted plane. Those recordings are retained but excluded, and three 1626 μs captures supply the fifth position. The 504 APS repeat-set depths

TABLE 1
Matched-support repeatability S_u in mm, with $n = 3$ acquisitions per command. $|R_u|$ is correlated row support, not repeat count.

Command (μs)	$\lvert R_u \rvert$	APS	Events
1569	21	3.200	3.321
1590	14	3.040	0.463
1602	16	3.527	3.394
1626	8	0.031	0.641
1647	8	0.682	1.127

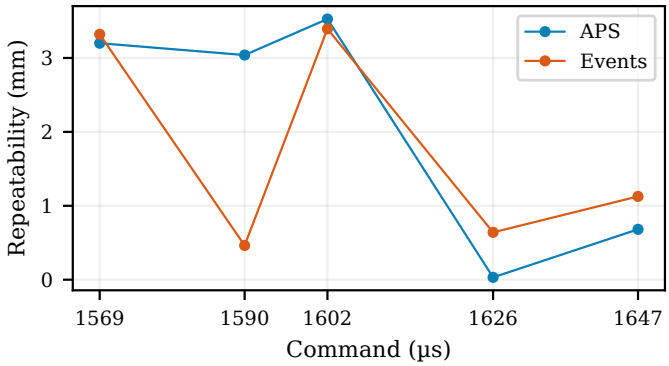


Fig. 10. Precision on the same rows in both methods and all three trials. Lines connect tested commands only and do not imply calibrated intermediate planes.

outside training ranges were retained rather than removed to improve the metric.

5.4 Unknown-filter configuration test

With the unknown filter, the camera signal fell from 27.135 ± 0.056 digital levels ($n = 2$) to 11.657 ± 0.088 ($n = 3$), a ratio of 0.4296 (sample SD across independent acquisitions, Fig. 11). Median stripe displacement remained approximately -0.94 to -0.96 pixels, so the ratio is not optical transmittance. APS retained stripe candidates in 62–64 rows while the event pipeline found 0–1. This is a negative result for our detector settings, not a general low-light limit of event cameras. Only two conditions were tested, so there is no multi-level attenuation curve or weakest-usable level.

5.5 Static timing prediction

Later ON markers have prediction RMS 25.74 ms and maximum absolute residual 47.08 ms, and all OFF markers have RMS 27.00 ms and maximum 47.55 ms (Fig. 12). This does not isolate clock drift from optical/event-selection effects, but it leaves the mapping insufficiently validated for assigning moving planes to events. No measured depth-error tolerance links these residuals to a safe sweep speed. Static held-plane reconstruction does not need that mapping.

6 Limitations and Pitfalls

Continuous reconstruction remains unvalidated. The pitch's main experiment needed each event matched to the laser plane at that instant. Our camera-to-controller time map left residuals up to 47.55 ms (Fig. 12), without a validated moving-plane error bound, so we used the pitch's

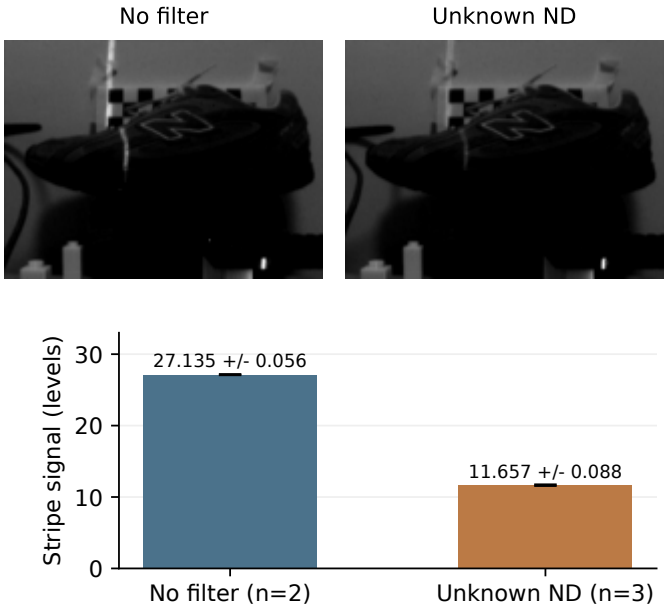


Fig. 11. Unknown-filter test at $1626\,\mu s$, 35 ms APS: actual median images above, mean stripe signal ± 1 sample SD below. No filter: APS rows 78/78, event rows 20/18 ($n = 2$); ND: APS 62/62/64, events 0/1/0 ($n = 3$). The roughly 1-pixel residual shift remains; the ratio is not optical transmission.

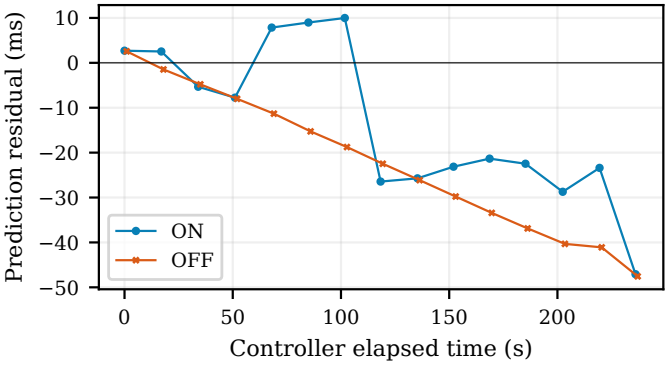


Fig. 12. Static optical-edge predictions after fitting the first five ON markers. Subsequent ON and all OFF markers are excluded from the fit. Residuals are not absolute timing uncertainty or a continuous-depth validation.

stop-and-pulse fallback. The speed study depends on this and was not run.

Events under attenuation. With one filter the event detector kept 0–1 rows while the image method kept 62–64 (Fig. 11). A reduced brightness step may suppress threshold crossings, but the optical signal, spatial shift and detector selection were not isolated, so this mechanism was not established. Detector settings were fixed after development and camera biases were not tuned. Earlier handling produced shifts of approximately -2 and $+14$ pixels; the retry still showed about -0.95 pixels. Only one unknown filter was tested.

No independent geometry. The board reference shares the camera model and the reported 18 mm print scale, so the 8.63 mm figure is held-out evidence, not absolute accuracy. We did not scan an independently measured object, such as a stepped block checked with calipers.

Position-dependent calibration. Each command has its own fitted plane. Two commands had only two usable board poses, and per-command error ranges from 0.88 to 17.03 mm. Planes are not interpolated between commands.

Resolution and coverage. The 240×180 sensor and 20 profiles sample only 6.85% (images) and 3.49% (events) of the visible shoe. This is image coverage, not 3D completeness.

Hardware substitutions. We used an Uno R3-type controller (10-bit ADC) instead of the planned R4, and a line module recorded as Class 3, driven from a digital pin, instead of the planned Class 1 module.

Pitfalls for next year. Use a fixed filter holder. Only one program can own the serial port at a time. Verify positional operation and feedback before mounting a replacement servo. Measure the printed board scale rather than assuming it.

Reproducibility. Code, CAD, build photographs, numerical summaries, calibration models and a coverage example are public at https://github.com/AryanRai/mech5720-2026-event-scanner. Running `python examples/coverage/regenerate.py` with the documented offline dependencies regenerates Fig. 7 from included coordinates, photo and mask, giving 939/13,715 APS and 478/13,715 event bins. A checksum script verifies retained evidence. The complete native archive is not publicly released, so all raw-data results cannot yet be reproduced from the repository alone. Original software is MIT-licensed, with third-party notices retained; final as-built specifications remain incomplete.

7 Conclusions and Future Work

The scanner produces partial 3D profiles from APS and static events. Reserved-board RMS is 8.63 mm with frozen calibration; matched-row results (events 8.07 mm, APS 8.42 mm) establish no advantage. The unknown-filter test exposes a detector failure. Continuous reconstruction remains unvalidated, motivating the static fallback.

With two more months, priorities are repeated tests on independently measured boards/steps; known attenuation levels in a fixed holder; and a bounded reference surface for physical completeness. Held-out physical timing and feedback-to-plane validation must precede continuous reconstruction and paired speed trials. Synchronisation wiring requires verified electrical compatibility. Measured servo limits would determine whether a faster actuator is justified.

Acknowledgments and Attribution

The team acknowledges the teaching staff, lab methods, cited work, OpenCV and iniVation tooling. OpenAI Codex assisted with software, analysis and drafting. Claude (Anthropic) assisted with the controller firmware, host logger, control dashboard, wiring diagram and talk animations. Photographs and measurement plots show retained hardware and data; CAD and the triangulation schematic show intended design and geometry. Final responsibility for reviewing the submission remains with the authors.

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